MALLA REDDY UNIVERSITY

DESIGN AND DEVELOPMENT INTELLIGENT SYSTEM FOR DEAF PERSONS

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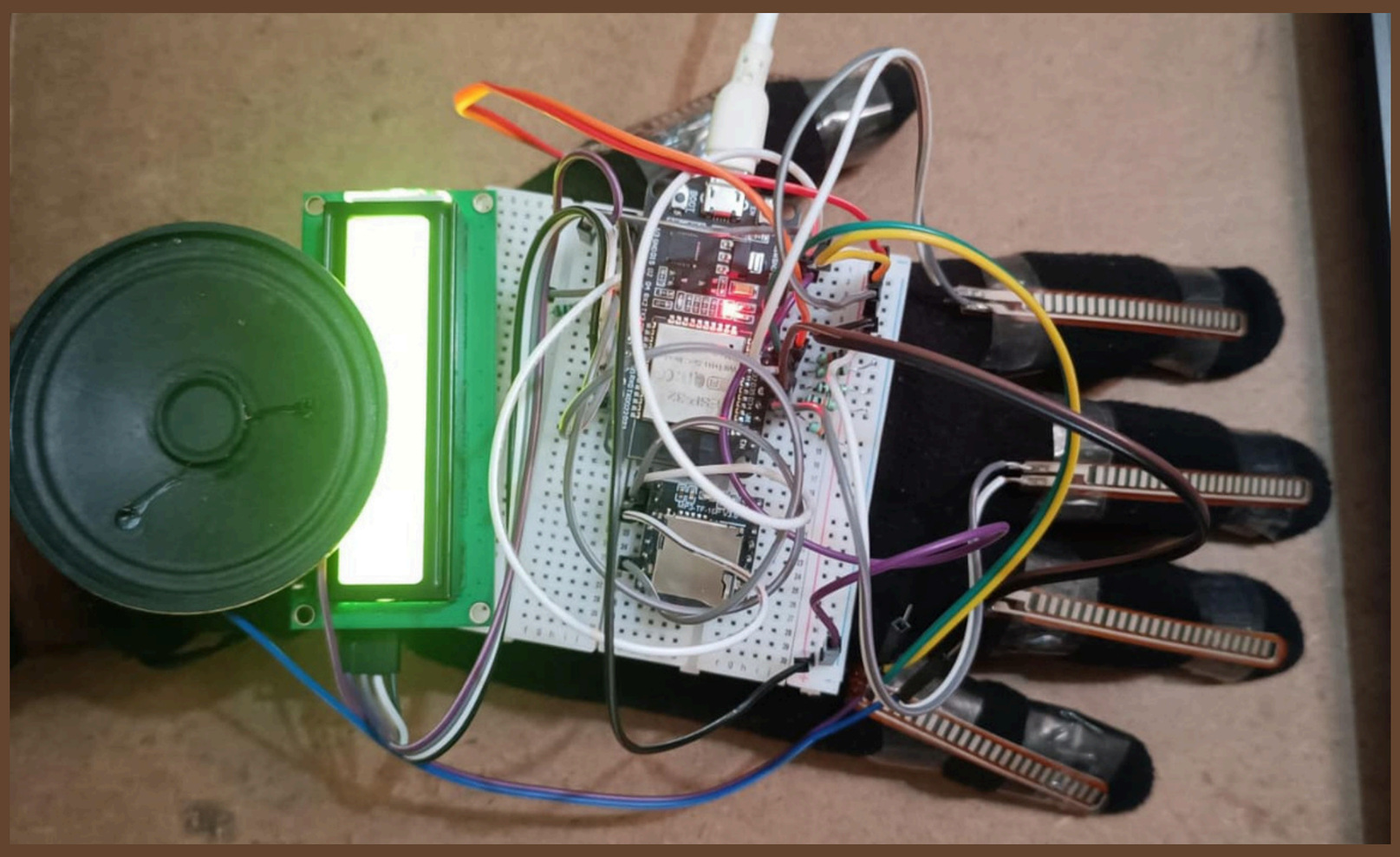
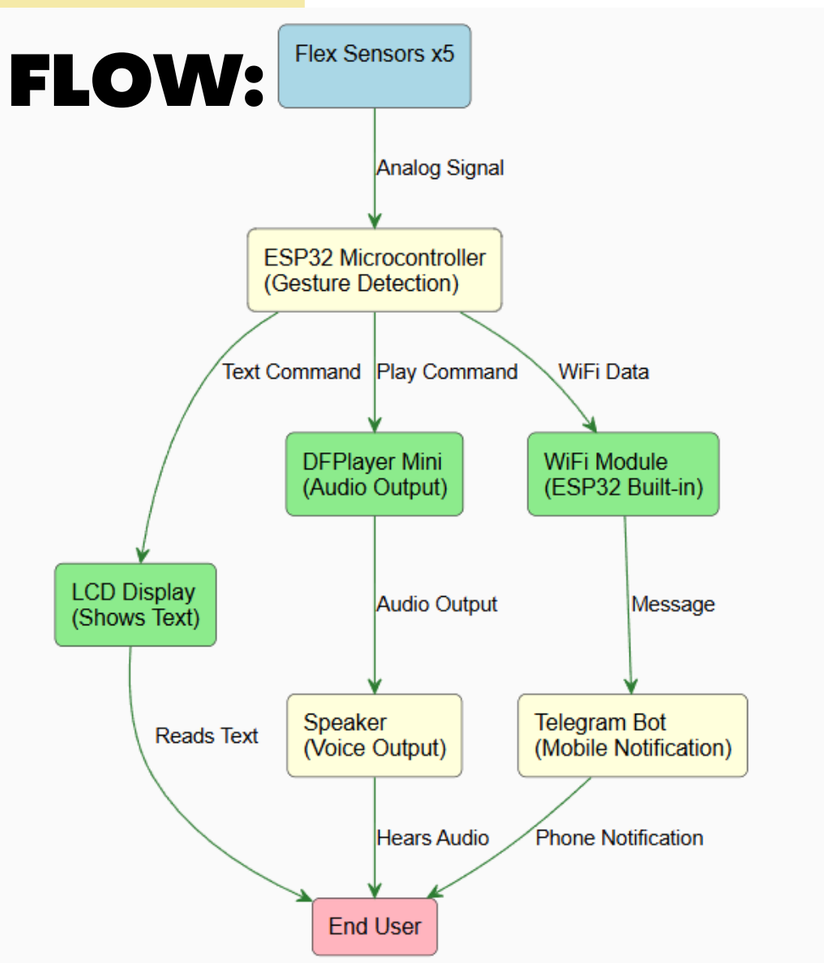
INTRODUCTION

Speech and hearing-impaired individuals face major challenges in daily communication. Existing solutions are expensive and inaccessible to common people.
This project proposes a Smart Sign Language Glove that detects hand gestures using flex sensors and converts them into speech, text, and mobile notifications instantly.
Built on ESP32 microcontroller with WiFi capability, the glove provides real-time output through an LCD display, speaker, and Telegram bot — making communication simple, affordable, and effective for deaf and mute individuals.

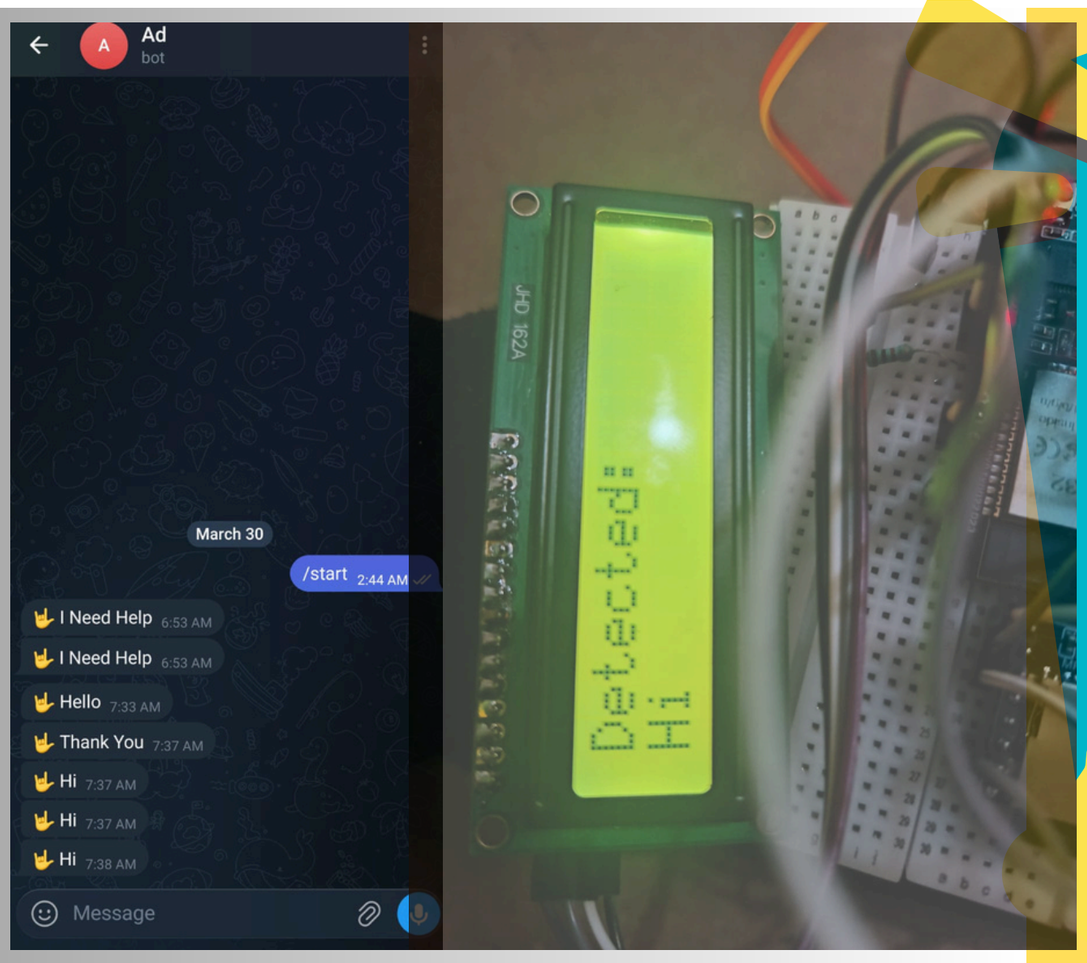


DATA FLOW:



ADVANTAGES:

- Multi-Output System — Simultaneously produces audio, visual display, and mobile notification for better communication.
- Wireless Communication — ESP32 WiFi sends real-time Telegram alerts to caregivers without any additional hardware.
- Easy to Use — Simple wearable glove design requires no technical knowledge to operate.
- Portable and Wearable — Compact design allows users to carry and use it anywhere in daily life.
- Real-Time Response — Instant gesture detection and output with minimal delay for smooth communication.
- Customizable — Gestures, messages and audio can be easily modified based on user requirements.
- No Internet Dependency for Local Output — LCD and speaker work locally even without WiFi connection



CONCLUSION:

THE SMART SIGN LANGUAGE INTERPRETATION GLOVE SUCCESSFULLY DEMONSTRATES A REAL-TIME ASSISTIVE COMMUNICATION SYSTEM FOR SPEECH AND HEARING-IMPAIRED INDIVIDUALS. BY INTEGRATING FLEX SENSORS, ESP32 MICROCONTROLLER, LCD DISPLAY, AUDIO OUTPUT, AND TELEGRAM NOTIFICATIONS, THE SYSTEM PROVIDES A COMPLETE MULTI-CHANNEL COMMUNICATION SOLUTION IN A SINGLE WEARABLE DEVICE.
THE PROJECT PROVES THAT ADVANCED ASSISTIVE TECHNOLOGY DOES NOT NEED TO BE EXPENSIVE OR COMPLEX. USING AFFORDABLE AND EASILY AVAILABLE COMPONENTS, THIS GLOVE BRIDGES THE COMMUNICATION GAP BETWEEN SPEECH-IMPAIRED INDIVIDUALS AND THE GENERAL PUBLIC IN AN EFFECTIVE AND PRACTICAL MANNER.
THE SYSTEM WAS SUCCESSFULLY TESTED FOR REAL-TIME GESTURE DETECTION, AUDIO OUTPUT, VISUAL DISPLAY, AND WIRELESS MOBILE NOTIFICATIONS — DEMONSTRATING RELIABLE PERFORMANCE ACROSS ALL OUTPUT CHANNELS SIMULTANEOUSLY.